Title

Student 1, Student 2, Student 3, Student 4, Student 5, Jorge A. Lizarraga, Graciela Baca, Luis Enrique González-Jiménez and Luis F. Luque-Vega

***Abstract*—The abstract is a single paragraph of 150–250 words, written last, once results already exist. It must be an objective synthesis of the whole paper, without stating conclusions that are not supported in the body of the text. Even though it has no visible subheadings, draft it following this internal structure:**
(1) Context: what question or need motivates the work?
(2) Methods: what was done to address it? (approach, materials, main methods, in 1-2 lines).
(3) Results: what was found? (the main findings, with concrete data if possible).
(4) Conclusion: what does that finding imply?
Avoid mathematical or chemical formulas inside the abstract.

***Index Terms*—keyword 1, keyword 2, keyword 3 (3 to 6 terms specific to the topic of the paper, yet reasonably common within the discipline – they help other researchers find the work).**

I. Introduction

THE introduction should place the work in a broad context and explain why it matters. It should cover, in this order:

- The problem or question that motivates the work, explained so that someone outside the specific field can understand it.
- The **motivation**: why this problem matters, who is affected by it, and what happens if it is not addressed.
- A brief preview that related work is reviewed next (the actual review goes in the Related Work subsection below).
- The gap, limitation, or open controversy that the reviewed related work leaves unresolved, and that this paper addresses.
- The concrete objective of the work.
- A preview of the main conclusion (one or two sentences).

Citation example: citing a source [1]. To cite several sources at once: [1], [2]. A closing paragraph that "summarizes the structure of the paper" is not required unless your journal/course explicitly asks for it.

A. Related Work

Summarize the state of the art relevant to this paper: similar robots or devices, similar methodologies, or comparable solutions already reported in the literature. For each relevant reference, briefly state what it solved and what it left open, citing the source with \cite{}. Close the subsection by clarifying how this work differs from, extends, or builds on that prior work – this is what justifies the contribution claimed in the objective above.

TABLE I
ESP32-C3 TECHNICAL SPECIFICATIONS AND COMPLIANCE ASSESSMENT.

Component	Parameter	Value	Source
ESP32-C3 (Espressif Systems, v2.4)	Supply Voltage (VDDA, VDD3P3, VDD3P3_RTC)	3.0–3.6 V; 3.3 V typical	Datasheet v2.4, Section 5.2, Table 5-2, p. 54
ESP32-C3	Cumulative Supply Current (IVDD)	0.5 A min.	Datasheet v2.4, Section 5.2, Table 5-2, p. 54
ESP32-C3	Peak Wi-Fi Transmission Current	276–335 mA, depending on the operating mode	Datasheet v2.4, Section 5.6.1, Table 5-7, p. 56
ESP32-C3	VIH: High-Level Input Voltage	$0.75 \times$ VDD min.	Datasheet v2.4, Section 5.4, Table 5-4, p. 55
ESP32-C3	VIL: Low-Level Input Voltage	-0.3 to $0.25 \times$ VDD V	Datasheet v2.4, Section 5.4, Table 5-4, p. 55
ESP32-C3	Strapping Pins / Boot Configuration Pins	GPIO2, GPIO8, and GPIO9. Their logic levels are sampled during reset to determine the boot mode.	Datasheet v2.4, Section 3, Boot Configuration, p. 30
ESP32-C3	GPIO8 and GPIO9 Boot-Time Levels	SPI Boot: GPIO9 = 1. Joint Download Boot: GPIO8 = 1 and GPIO9 = 0. The GPIO8 = 0, GPIO9 = 0 combination is not listed as a valid boot mode in Table 3-3.	Datasheet v2.4, Section 3.1, Table 3-3, p. 31
ESP32-C3	Wi-Fi	2.4 GHz, IEEE 802.11 b/g/n	Datasheet v2.4, Features section, p. 3
ESP32-C3	Bluetooth	Bluetooth 5	Datasheet v2.4, Features section, p. 3
ESP32-C3	Operating Temperature	56 °C	Datasheet v2.4, Section 1.2, Table 1-1, p. 12

II. Materials and Methods

This section must have enough detail for someone else to replicate the work. Include, as applicable:

- The materials, components, equipment, datasets, or tools used, with relevant specifications (well-established methods do not need to be justified in detail; citing them is enough).
- The methods or procedures applied, described with enough detail to be reproducible. If a method is new, describe it in detail; if it is a known method, a citation plus a brief description is enough.

The authors are with the Department of Technological and Industrial Processes, ITESO, Tlaquepaque 45604, Jalisco, Mexico.

TABLE II
SERVOMOTOR MG90S TECHNICAL SPECIFICATIONS AND COMPLIANCE ASSESSMENT.

Component	Parameter	Value	Source
MG90s	Supply and operating Voltage	4.8 V – 6.0 V	MG90S datasheet, Tower Pro, p.2
MG90s	Stall Torque	1.8 kgf·cm at 4.8V, 2.2 kgf·cm at 6V	MG90S datasheet, Tower Pro, p.1
MG90s	Operating Speed	0.1 s/60° at 4.8V, 0.08 s/60° at 6V	MG90S datasheet, Tower Pro, p.1
MG90s	Control Signal	PWM, orange wire; Vcc red; GND brown; 20 ms (50 Hz) period, 1–2 ms duty cycle	MG90S datasheet, Tower Pro, p.2
MG90s	Rotation Range	approximately 180 degrees (90° in each direction)	MG90S datasheet, Tower Pro, p.2
MG90s	Dead Band Width	5 μs	MG90S datasheet, Tower Pro, p.2
MG90s	Operating Temperature	Not specified in datasheet	N/A
MG90s	Dimension & Weight	22.5 × 12 × 35.5 mm approx.; 13.4 g	MG90S datasheet, Tower Pro, p.1

- If the work uses public data or generates its own dataset, state where it is available (or state that it will be provided during review).
- If the work involves human or animal subjects, or requires ethical approval, state the authority that granted it and the approval code. If not applicable, this can be omitted.
- If generative artificial intelligence was used (beyond superficial text editing) to generate text, data, or graphics, or to support the study design, data collection, or analysis, disclose it briefly here.

> Example of a long quotation or of a highlighted note within the text (use this "quote" block sparingly).

III. DESIGN AND IMPLEMENTATION OF THE TECHNOLOGICAL SOLUTION

This section shows how the methods and materials from the previous section were turned into the actual built solution. Include, as applicable:

- The overall system architecture (block diagram) showing how the different subsystems interact.
- Mechanical, electronic, and/or software design decisions, with enough detail to be reproducible.
- How the different subsystems were integrated into the complete solution.
- Configuration, calibration, or programming steps needed to bring the system into working operation.

Use figures (block diagrams, CAD renders, architecture diagrams, photographs of the assembled system) and tables to support this description, following the same figure/table conventions shown in the Results section below.

TABLE III
PCA9685 TECHNICAL SPECIFICATIONS AND COMPLIANCE ASSESSMENT.

Component	Parameter	Value	Source
PCA9685 (NXP Semiconductors, datasheet Rev. 4)	Supply Voltage (V_{DD})	2.3–5.5 V	Datasheet Rev. 4, Section 2, Table 14, p. 38
PCA9685	PWM Resolution	12 bits (4096 steps)	Datasheet Rev. 4, Section 1, p. 1
PCA9685	Number of PWM Channels	16 independent channels	Datasheet Rev. 4, Section 1, p. 1
PCA9685	Communication Interface	I^{2}C Fast-mode Plus	Datasheet Rev. 4, Section 2, p. 2
PCA9685	Maximum I^{2}C Frequency	Up to 1 MHz	Datasheet Rev. 4, Section 2, p. 2
PCA9685	PWM Frequency Range	24 Hz – 1526 Hz	Datasheet Rev. 4, Section 7.3.5, p. 25
PCA9685	Default PWM Frequency	Approximately 200 Hz	Datasheet Rev. 4, Table 8, p. 25
PCA9685	I^{2}C Address Configuration	A0–A5 pins, up to 62 devices	Datasheet Rev. 4, Section 7.1, p. 7
PCA9685	Output Current	25 mA per output	Datasheet Rev. 4, Table 13, p. 38
PCA9685	Total Output Current	400 mA maximum	Datasheet Rev. 4, Table 13, p. 38
PCA9685 Module (UNIT Electronics)	Output Ports	16 PWM outputs with V+, GND and PWM signal	UNIT Electronics product specification
PCA9685 Module (UNIT Electronics)	Logic Voltage	3–5 V	UNIT Electronics product specification
PCA9685 Module (UNIT Electronics)	Servo Power Supply (V+)	External power supply for connected actuators	UNIT Electronics product specification
PCA9685	Operating Temperature	-40 °C to $+85$ °C	Datasheet Rev. 4, Table 13, p. 38

Algorithm 1 SOFIA Module Inverse Kinematics

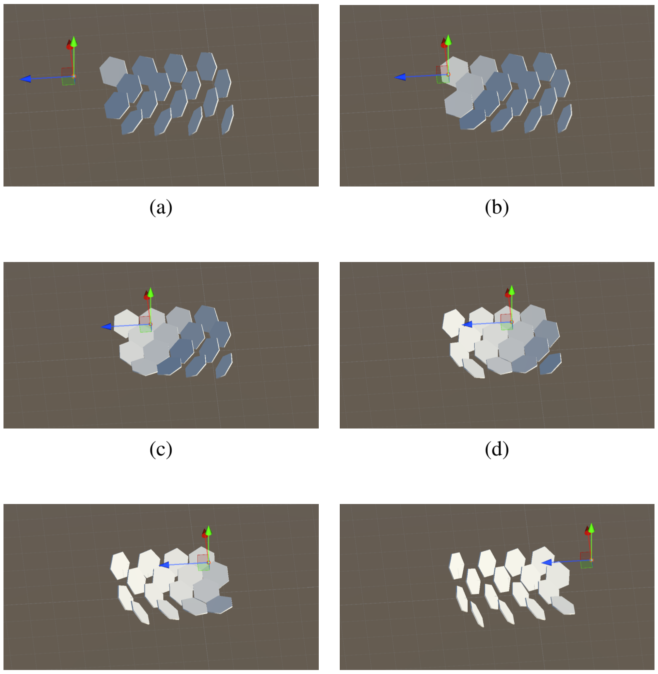
Require: Module base origin $\mathbf{r}_{0,2}$, target coordinate $\mathbf{p}$, joint angle bounds $[q_{1,\min}, q_{1,\max}]$ and $[q_{2,\min}, q_{2,\max}]$
Ensure: Joint angles q_1, q_2
1: $\mathbf{r}_{2,p} \leftarrow \mathbf{p} - \mathbf{r}_{0,2}$
2: $[u_x, u_y, u_z]^\top \leftarrow \mathbf{r}_{2,p}/\|\mathbf{r}_{2,p}\|$
3: $q_2 \leftarrow \arcsin(u_x)$
4: $q_1 \leftarrow \operatorname{atan2}(-u_y, u_z)$
5: $q_1 \leftarrow \textsc{ClampAngle}(q_1, q_{1,\min}, q_{1,\max})$
6: $q_2 \leftarrow \textsc{ClampAngle}(q_2, q_{2,\min}, q_{2,\max})$
7: **return** q_1, q_2

IV. RESULTS

A. Kinematic Tracking Simulation

To validate the analytical kinematic model, a dynamic tracking simulation of the 16-module SOFIA robotic honeycomb array was conducted in a virtual 3D environment. The goal was to evaluate multi-unit coordination when orienting toward a moving target coordinate $\mathbf{p}(t) \in \mathbb{R}^3$.

Algorithm 1 summarizes the inverse kinematics routine executed cyclically for each module m_i ($i \in \{1, \ldots, 16\}$) based on the unit base frame origin $\mathbf{r}_{0,2}^{(i)}$, solving for servo variables (q_1, q_2) while clamping values within the permissible physical limits of the MG90S actuators ($[-90°, 90°]$).



(a) (b) (c) (d) (e) (f)

Fig. 1. Kinematic tracking sequence of the 16-module SOFIA honeycomb array following a 3D target $\mathbf{p}(t)$. (a)–(c) Target boundary ingress and progressive orientation alignment across adjacent units. (d)–(f) Peak collective angular deflection and trajectory transit maintaining normal vector tracking under joint constraints.

Figure 1 presents the multi-body dynamic response across six key stages of the simulated tracking cycle: (a)

1) *Target Ingress:* As $\mathbf{p}(t)$ enters the workspace periphery, initial orientation is led by the closest boundary modules ($|q_1|, |q_2| < 15°$), while distal cells remain in their nominal vertical orientation.
2) *Cluster Propagation:* The continuous decrease in $|\mathbf{r}_{2,p}|$ triggers an inward cascade of servo commands across adjacent rows without phase lag.
3) *Intermediate Convergence:* The entire 16-element array forms a focused directional field, uniformly tilting the normal vectors $\hat{r}_{2,3}^{(i)}$ toward the approaching coordinate.
4) *Peak Deflection:* Central transit coincides with maximum angular excursion across the inner cluster, smoothly bounding within actuator limits without kinematic locking.
5) *Quadrant Handover:* As the target traverses into the opposite quadrant, neighboring units execute continuous angular redistribution.
6) *Terminal Alignment:* The full array converges into final steady-state tracking, verifying robust singularity-free coverage across the entire hemispherical workspace.

V. Discussion

This is where the results are interpreted, not just repeated. It should answer:

- What do the obtained results mean?
- How do they compare with the state of the art cited in the introduction? Do they confirm, contradict, or qualify previous work?
- What limitations does the work have (methodological, of scope, of validity)?
- What practical or theoretical implications does the finding have?
- What future work does this open up?

This section can be merged with Results if your journal/course prefers that (check the specific instructions), but keeping them separate is recommended to avoid mixing ”what was found” with ”what it means”.

VI. Conclusions

A brief synthesis (not a repetition of the abstract or of the discussion) of the answer to the question posed in the introduction, and of the concrete contribution of the work. This section is not mandatory if the discussion already provides that closure, but it is recommended when the paper is long or the discussion covered several topics.

Author Contributions

If the paper has several authors, briefly specify each person's individual contribution (e.g., conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing, visualization, supervision, project administration, funding acquisition). Authorship must be limited to those who contributed substantially to the reported work.

Data Availability

State where the data, code, or materials that support the reported results can be found (repository, link, DOI). If no new data were generated, or if there are privacy or confidentiality restrictions, state that as well.

Acknowledgment

Acknowledge people or institutions that supported the work but do not qualify as authors (technical or administrative support, in-kind donations, funding). If generative AI was used in preparing the manuscript, disclose it here with the tool name, version, and the specific use it was given.

Conflicts of Interest

Choose whichever of the following applies, or adapt it to your case:

• **No conflict**: state “The authors declare no conflicts of interest.”

• **Declared conflict**: identify and declare any personal circumstances or interests that could be perceived as inappropriately influencing the representation or interpretation of the reported results.

• **Role of funders**: state the role of any funders in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results. If there was no such role, state: “The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.”

References

[1] N. Apellido, “Título del artículo de ejemplo,” *Nombre de la revista*, vol. 1, pp. 1–10, 2026.

[2] N. Apellido and N. Apellido, “Título del trabajo presentado,” in *Nombre de la conferencia*, Ciudad, País, 2026, pp. 1–6.